

Foundations for programming languages

9: Answers for exercises

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Exercise 1

- (1) Obtain the free variables in $(\lambda z. z) w$ according to the definition of free variables.
- (2) Obtain the free variables in $(\lambda z. z y) ((\lambda z. z) w)$ according to the definition of free variables.

An answer

(1)

$$\begin{aligned} \text{FV}((\lambda z. z) w) &= \text{FV}((\lambda z. z)) \cup \text{FV}(w) \\ &= (\text{FV}(z) \setminus \{z\}) \cup \{w\} \\ &= (\{z\} \setminus \{z\}) \cup \{w\} \\ &= \{\} \cup \{w\} \\ &= \{w\} \end{aligned}$$

(2)

$$\begin{aligned} \text{FV}((\lambda z. z y) ((\lambda z. z) w)) &= \text{FV}(\lambda z. z y) \cup \text{FV}((\lambda z. z) w) \\ &= (\text{FV}(z y) \setminus \{z\}) \cup (\text{FV}(\lambda z. z) \cup \text{FV}(w)) \\ &= ((\text{FV}(z) \cup \text{FV}(y)) \setminus \{z\}) \cup ((\text{FV}(z) \setminus \{z\}) \cup \{w\}) \\ &= ((\{z\} \cup \{y\}) \setminus \{z\}) \cup ((\{z\} \setminus \{z\}) \cup \{w\}) \\ &= (\{z, y\} \setminus \{z\}) \cup (\{\} \cup \{w\}) \\ &= \{y\} \cup \{w\} \\ &= \{y, w\} \end{aligned}$$

Exercise 2

- (1) What does $(x\ y)\ [z/x]$ represent?
- (2) What does $(\lambda y. x\ y)\ [z/x]$ represent?
- (3) What does $(\lambda y. x\ y)\ [y/x]$ represent?
- (4) What does $(\lambda y. x\ y)\ [\lambda z. z\ y/x]$ represent?

Answers

(1)

$$\begin{aligned}(x \ y) \ [z/x] &= (x \ [z/x]) \ (y \ [z/x]) \\ &= z \ y\end{aligned}$$

(2)

$$\begin{aligned}(\lambda y. x \ y) \ [z/x] &= \lambda y. ((x \ y) \ [z/x]) \\ &= \lambda y. ((x \ [z/x]) \ (y \ [z/x])) \\ &= \lambda y. (z \ y) \quad (\text{The parentheses can be ommited.})\end{aligned}$$

(3)

$$\begin{aligned}(\lambda y. x \ y) \ [y/x] &= \lambda z. (((x \ y) \ [z/y]) \ [y/x]) \\ &= \lambda z. (((x \ [z/y]) \ (y \ [z/y])) \ [y/x]) \\ &= \lambda z. ((x \ z) \ [y/x]) \\ &= \lambda z. ((x \ [y/x]) \ (z \ [y/x])) \\ &= \lambda z. (y \ z) \quad (\text{The parentheses can be ommited.})\end{aligned}$$

Answers (cont.)

(4)

$$\begin{aligned}(\lambda y. x y) [\lambda z. z y/x] &= \lambda w. (((x y) [w/y]) [\lambda z. z y/x]) \\&= \lambda w. (((x [w/y]) (y [w/y])) [\lambda z. z y/x]) \\&= \lambda w. ((x w) [\lambda z. z y/x]) \\&= \lambda w. ((x [\lambda z. z y/x]) (w [\lambda z. z y/x])) \\&= \lambda w. ((\lambda z. z y) w)\end{aligned}$$

(The outer parentheses can be omitted.)

Exercise 3

- (1) Beta reduce once $(\lambda x. x y) (\lambda z. z)$.
- (2) Beta reduce once $(\lambda x. (\lambda y. x y)) (\lambda z. y z)$.

Answers

(1)

$$\begin{aligned}(\lambda x. x y) (\lambda z. z) &\rightarrow (x y) [\lambda z. z/x] \\&= (x [\lambda z. z/x]) (y [\lambda z. z/x]) \\&= (\lambda z. z) y \\&\text{(This can further be beta reduced.)}\end{aligned}$$

(2)

$$\begin{aligned}(\lambda x. (\lambda y. x y)) (\lambda z. y z) &\rightarrow (\lambda y. x y) [\lambda z. y z/x] \\&= \lambda z. (((x y) [z/y]) [\lambda z. y z/x]) \\&= \lambda z. (((x [z/y]) (y [z/y])) [\lambda z. y z/x]) \\&= \lambda z. ((x z) [\lambda z. y z/x]) \\&= \lambda z. ((x [\lambda z. y z/x]) (z [\lambda z. y z/x])) \\&= \lambda z. ((\lambda z. y z) z) \\&\text{(The outer parentheses can be omitted.)} \\&\text{(This can further be beta reduced.)}\end{aligned}$$

Exercise 4

A lambda expression $(\lambda x. \lambda y. x y) (\lambda z. z) w$ can be transformed to w by applying beta reductions. Write the each step of beta reductions.

An answer

$$\begin{aligned} & (\lambda x. \lambda y. x y) (\lambda z. z) w \\ & \rightarrow (\lambda y. (\lambda z. z) y) w \\ & \rightarrow (\lambda y. y) w \\ & \rightarrow w \end{aligned}$$

or

$$\begin{aligned} & (\lambda x. \lambda y. x y) (\lambda z. z) w \\ & \rightarrow (\lambda y. (\lambda z. z) y) w \\ & \rightarrow (\lambda z. z) w \\ & \rightarrow w \end{aligned}$$

(Note that we omit the description in the substitution notation.)

Exercise 5

Beta reduce the lambda expression $(\lambda x. \lambda y. x y) (\lambda x. x y) w$ until obtaining a lambda expression that can not be beta reduced.

In this example, there are two sequences of beta reductions. Show both of them.

An answer

$$\begin{aligned} & (\lambda x. \lambda y. x y) (\lambda x. x y) w \\ & \rightarrow (\lambda z. (\lambda x. x y) z) w \\ & \rightarrow (\lambda z. z y) w \\ & \rightarrow w y \end{aligned}$$

or

$$\begin{aligned} & (\lambda x. \lambda y. x y) (\lambda x. x y) w \\ & \rightarrow (\lambda z. (\lambda x. x y) z) w \\ & \rightarrow (\lambda x. x y) w \\ & \rightarrow w y \end{aligned}$$

(Note that we omit the description in the substitution notation.)